Stabilization Policy with Rational Expectations

Macroeconomics B
Lecture 8 (Chapters 21–22)

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Where We Are

You have built the short-run model in three blocks:

- **Aggregate demand (AD):** output demand falls when the relevant real interest rate rises.
- **Aggregate supply (AS):** inflation depends on expected inflation, the output gap, and supply shocks.
- **Aggregate supply–aggregate demand (AS–AD) dynamics:** shocks move curves, and expectations determine persistence.

Today we replace backward-looking expectations with forward-looking, model-consistent expectations.

Question: when can stabilization policy still move output and inflation if people understand the policy rule?

Roadmap

1. Why expectations are part of the transmission mechanism
2. From static expectations to rational expectations
3. Why systematic policy may become ineffective
4. Why timing and information restore policy effectiveness
5. Optimal stabilization and the policy trade-off
6. Credibility, time inconsistency, and inflation bias
7. Measurement errors and cautious policy
8. Deviating from pure rational expectations

Learning Goals

After this lecture, you will be able to:

- define rational expectations before using them in the AS curve,
- explain why static expectations can imply predictable forecast errors,
- state the policy ineffectiveness proposition and its intuition,
- distinguish policy based on old information from policy based on current information,
- explain the demand-shock versus supply-shock stabilization trade-off,
- use credibility to understand inflation bias,
- explain why measurement errors and lags make aggressive stabilization risky,
- interpret mixed expectations as a deviation from pure rational expectations.

Why Expectations Are a Policy Channel

Policy affects private decisions not only through today's interest rate, but also through expected future policy.

Policy communication	Private-sector expectation	Immediate effect
Lower inflation target	lower future inflation	wage and price setting changes
Forward guidance	lower future short rates	long rates may fall today
Credible independent central bank	low inflation is expected	less need for costly disinflation
Weak credibility	promises may not be believed	policy has less traction

The mechanism is: expected policy $\Rightarrow$ expected inflation and interest rates $\Rightarrow$ consumption, investment, output, and inflation.

Definitions Used Today

- y_t: output in period t; $\bar{y}$: structural output.
- $\widehat{y}_t \equiv y_t - \bar{y}$: output gap.
- π_t: inflation; π^*: inflation target.
- $\widehat{\pi}_t \equiv \pi_t - \pi^*$: inflation gap.
- $\pi^e_{t,t-1}$: inflation expected for period t, formed in period $t-1$.
- v_t: demand shock; s_t: supply shock; $\widehat{\rho}_t$: interest-rate spread shock.

The short-run AS curve is

$$\pi_t = \pi^e_{t,t-1} + \gamma \widehat{y}_t + s_t,$$

where $\gamma > 0$ measures how strongly the output gap raises inflation.

Static Expectations: Useful, but Too Mechanical

Static expectations assume that expected inflation equals last period's inflation:

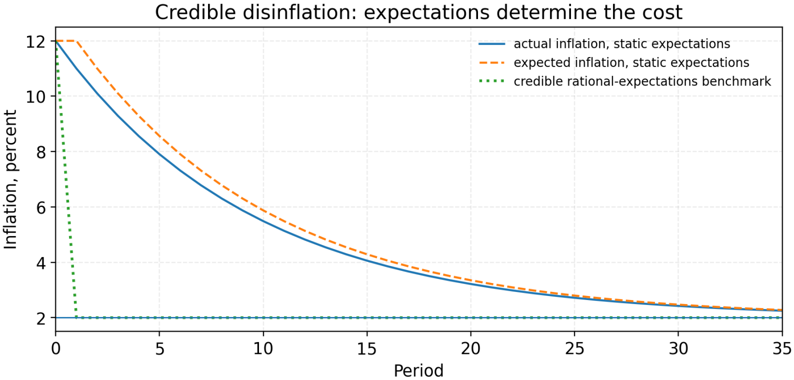
$$\pi^e_{t,t-1} = \pi_{t-1}.$$

This helps explain persistence:

- high inflation today becomes expected inflation tomorrow,
- the AS curve shifts only gradually,
- output and inflation return slowly after a shock.

But static expectations can also imply repeated, predictable mistakes. If a credible policy change is announced, why would agents ignore it and keep forecasting only from yesterday?

A Credible Disinflation: Static Expectations versus Forward Looking



Source: Own simulation, generated by `generate_18_figures.py`.
If the new target is credible, inflation expectations can move immediately; with static expectations, disinflation is unnecessarily slow.

Rational Expectations

Rational expectations (RE) means that subjective forecasts equal the model's best forecast, conditional on available information.

$$X^e_{t,t-1} = E(X_t \mid I_{t-1}).$$

- X_t: any variable to be forecast.
- I_{t-1}: information available when the forecast is made.
- The model includes economic relationships, policy rules, and shock distributions.

RE does **not** mean perfect foresight. Shocks realized after the forecast can still create forecast errors. The point is that these errors are not systematic.

Timing: Why Information Matters

The policy result depends on who knows what, and when.

Step	Private sector	Central bank
$t-1$	forms $\pi^e_{t,t-1}$ and wage/price plans	observes the same old information
t	shocks $v_t, s_t, \widehat{\rho}_t$ are realized	may or may not observe current conditions
t policy	reacts according to a rule	either uses expected gaps or actual gaps
t outcome	output and inflation are realized	policy effect is observed

If the central bank and private sector have the same information, systematic policy is anticipated. If the central bank observes shocks later, it can react to information not used in wage and price setting.

Case 1: Policy Uses Only Old Information

Suppose the central bank sets the policy rate using expected current gaps, not the realized gaps:

$$\text{policy reacts to } \pi^e_{t,t-1} - \pi^* \quad \text{and} \quad y^e_{t,t-1} - \bar{y}.$$

Under rational expectations, private agents know the rule and understand how it affects the economy.

In the absence of expected shocks, the best forecasts are

$$E(y_t \mid I_{t-1}) = \bar{y}, \qquad E(\pi_t \mid I_{t-1}) = \pi^*.$$

Therefore the systematic part of the policy rule is already priced into expectations.

Policy Ineffectiveness Proposition

The **policy ineffectiveness proposition (PIP)** says:

> If expectations are rational and policy uses no information that private agents lack, then predictable monetary policy does not systematically move output.

The realized variables still move because shocks occur:

$$\widehat{y}_t = v_t - \alpha_2 \widehat{\rho}_t, \qquad \widehat{\pi}_t = \gamma v_t + s_t - \alpha_2 \gamma \widehat{\rho}_t.$$

The Taylor-rule parameters do not appear in these impact outcomes.

▸ Appendix derivation

The Intuition Behind Policy Ineffectiveness

Start from the AS curve:

$$\pi_t = \pi^e_{t,t-1} + \gamma \widehat{y}_t + s_t.$$

A non-zero output gap requires either:

- a supply shock s_t, or
- surprise inflation, meaning $\pi_t - \pi^e_{t,t-1} \neq 0$.

Under rational expectations and equal information, a known policy rule cannot create systematic surprise inflation.

The central bank can still affect the economy through unexpected policy mistakes or through responses to new information. It cannot fool agents in a predictable way.

Case 2: Policy Uses Current Information

Now suppose the central bank observes current output and inflation before setting the policy rate:

$$\text{policy reacts to } \pi_t - \pi^* \quad \text{and} \quad y_t - \bar{y}.$$

Private agents formed $\pi^e_{t,t-1}$ before current shocks were known.
This information advantage matters:

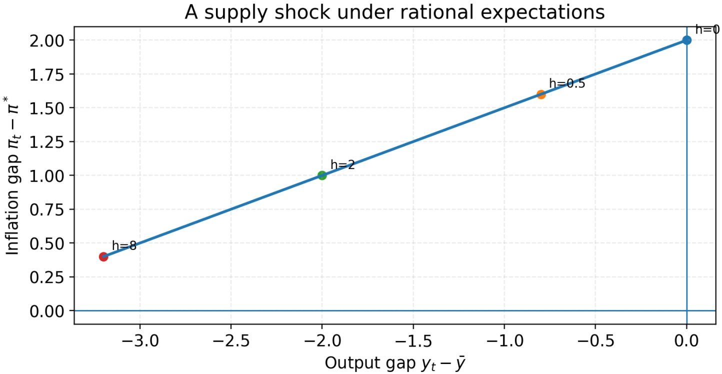
- a current demand shock can be offset by moving the real interest rate,
- a current supply shock creates an inflation-output trade-off,
- systematic policy is effective because it responds to information not used in expectations.

Demand and Supply Shocks under Current-Information Policy

Shock type	What happens without policy?	What policy can do
Demand shock v_t	output and inflation move in the same direction	stabilize both by offsetting demand
Supply shock s_t	inflation rises while output falls	choose how much inflation to reduce and how much output loss to accept
Spread shock $\widehat{\rho}_t$	borrowing costs rise and demand falls	partly offset if the shock is observed

Demand shocks do not create a fundamental stabilization trade-off. Supply shocks do, because they push inflation and output in opposite directions.

A Supply Shock: The Trade-off Is Real



Source: Own simulation, generated by `generate_18_figures.py`.
A stronger inflation response lowers the inflation gap after a supply shock, but it does so by accepting a larger negative output gap.

Optimal Stabilization: What Is Being Minimized?

A standard social loss function is

$$SL_t = (y_t - \bar{y})^2 + \kappa(\pi_t - \pi^*)^2, \qquad \kappa > 0.$$

- The first term penalizes output instability.
- The second term penalizes inflation instability.
- κ measures the relative weight on inflation stability.

The policy choice is not simply "fight inflation" or "protect output". It is to choose the point on the feasible trade-off that creates the smallest total loss.

For detailed algebra, see the appendix. Optimal response

What Rational Expectations Do to Persistence

In the static-expectations AS–AD model, persistence comes from

$$\pi^e_{t,t-1} = \pi_{t-1}.$$

High inflation feeds directly into the next period's expectations.

Under pure rational expectations, predictable adjustment is already anticipated. With one-period contracts and no serially correlated shocks:

- there is no automatic inflation inertia from yesterday's inflation,
- short-run movements come from shocks and information timing,
- persistence must come from persistent shocks, contracts, learning, or backward-looking elements.

Lucas Critique

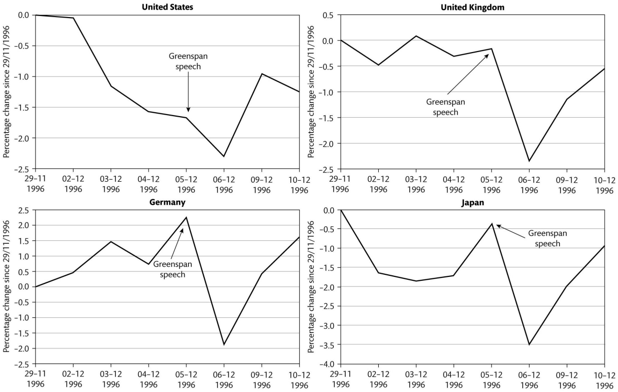
The **Lucas critique** says that reduced-form empirical relationships may change when policy changes.

Example:

- Under one policy regime, high money growth may be associated with high output.
- If the central bank systematically exploits that relationship, people revise inflation expectations.
- The old relationship no longer predicts the effects of the new policy.

The lesson is not that data are useless. The lesson is that policy analysis needs behavioral relationships that are stable when policy rules change.

Announcements Can Move Markets Immediately



Source: Sørensen and Whitta-Jacobsen, Chapter 21.
Asset prices can respond as soon as agents revise expected future policy, even before current output or inflation has changed.

Credibility and the Policy Problem

Credibility means that private agents believe the announced policy rule.

Why it matters:

- If low inflation is credible, expected inflation stays low.
- If expected inflation is low, wage and price setting starts from a low baseline.
- If credibility is weak, disinflation requires larger output losses.

Credibility is not just about reputation. It changes the AS curve through expected inflation.

Rules versus Discretion

A **policy rule** is a preannounced conditional response. A Taylor-type rule is one example.

Discretion means policy is chosen freely after observing the situation.

Criterion	Rule	Discretion
Flexibility	lower	higher
Credibility	easier to build	harder to build
Expectation management	clearer	less predictable
Risk	too rigid	time inconsistency

The central question is whether the flexibility of discretion is worth the credibility cost.

Inflation Bias: The Time-Inconsistency Logic

A government or central bank may prefer output above the structural level, for example because unemployment is politically costly.

The temptation is:

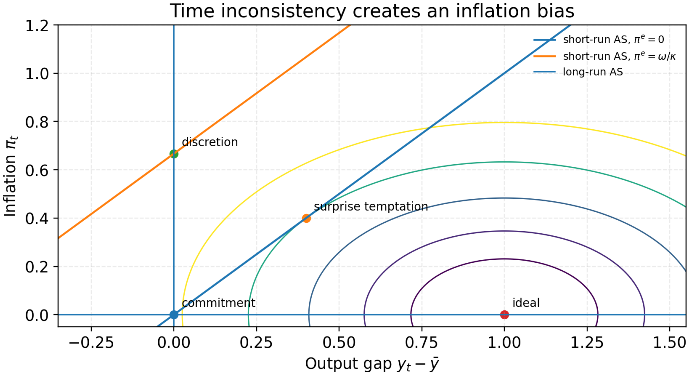
$$\text{promise low inflation} \quad \Rightarrow \quad \text{create surprise inflation} \quad \Rightarrow \quad \text{temporarily higher output.}$$

With rational expectations, agents understand this temptation.

- They expect higher inflation.
- Surprise inflation disappears.
- Output returns to its structural level.
- Inflation remains too high.

This is the inflation-bias problem.

Inflation Bias in One Picture



Source: Own illustration based on the Barro–Gordon model, generated by `generate_18_figures.py`.
Discretion removes the output gain but leaves positive inflation; commitment can anchor expectations and avoid the bias.

How Institutions Reduce Inflation Bias

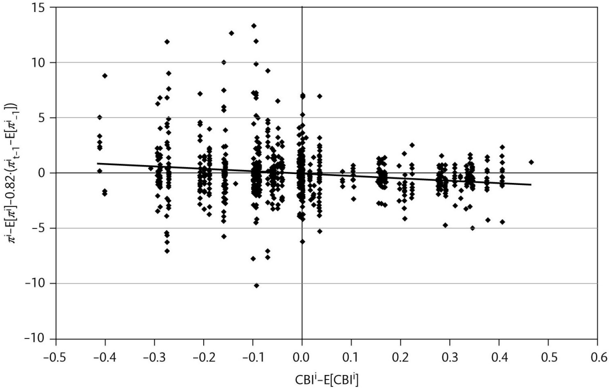
The policy problem is not just technical. It is institutional.

Common solutions:

- **Commitment:** make future policy harder to reverse.
- **Reputation:** repeated behavior makes promises more believable.
- **Delegation:** assign monetary policy to an independent, inflation-focused central bank.
- **Transparency:** publish forecasts, minutes, and reaction functions.

Each solution works by changing expectations before private wage and price setting takes place.

Central Bank Independence and Inflation



Source: Sørensen and Whitta-Jacobsen, Chapter 22.
A more independent central bank is associated with lower inflation on average, consistent with an institutional solution to inflation bias.

Why Not React Aggressively to Everything?

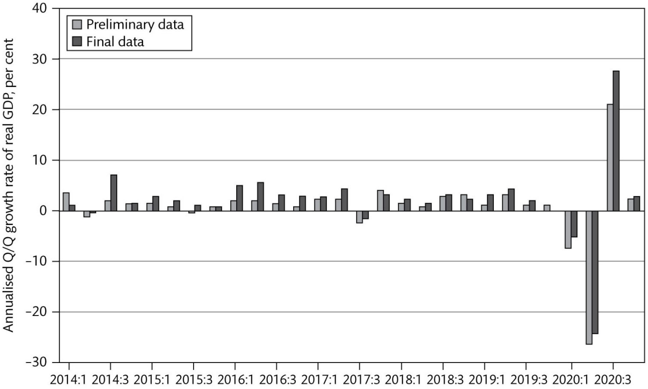
Aggressive stabilization is attractive in the simple model, but real-time policy uses imperfect data.

Three practical problems:

- **Measurement error:** gross domestic product (GDP), inflation, and the output gap are measured with noise.
- **Revisions:** preliminary data can change substantially later.
- **Lags:** policy decisions affect the economy with delay.

If policy reacts strongly to noisy data, it may stabilize the noise instead of the economy.

Real-Time Data Can Be Misleading



Source: Sørensen and Whitta-Jacobsen, Chapter 22.
Preliminary GDP data can differ from final data, so an aggressive real-time response may be based on the wrong signal.

Deviating from Pure Rational Expectations

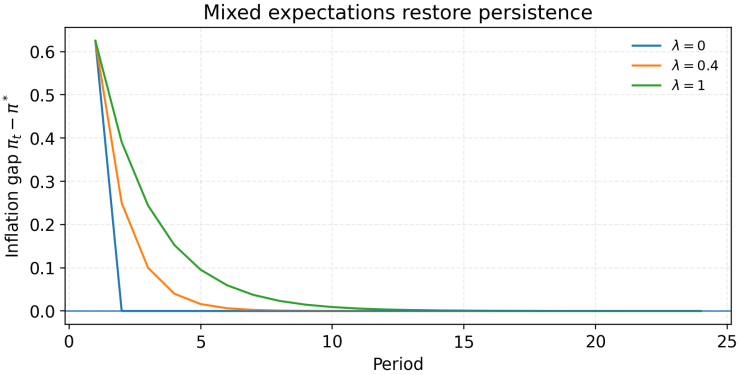
A simple mixed-expectations rule is

$$\pi^e_{t,t-1} = \lambda \pi_{t-1} + (1-\lambda) E(\pi_t \mid I_{t-1}), \qquad 0 \le \lambda \le 1.$$

- $\lambda = 0$: pure rational expectations.
- $\lambda = 1$: fully static expectations.
- $0 < \lambda < 1$: expectations are partly backward-looking.

Pure rational expectations is a benchmark. This small deviation is enough to bring back persistence after shocks.

Mixed Expectations Restore Persistence



Source: Own simulation, generated by `generate_18_figures.py`.
The more backward-looking expectations are, the longer inflation remains away from target after a temporary supply shock.

Takeaways

- Rational expectations means model-consistent forecasting, not perfect foresight.
- If policy uses no information that private agents lack, predictable policy cannot systematically create output surprises.
- If the central bank observes current shocks before acting, stabilization policy can matter.
- Demand shocks can be offset without a fundamental inflation-output trade-off.
- Supply shocks force a choice between inflation stabilization and output stabilization.
- Credibility matters because it moves expected inflation and therefore the AS curve.
- Data uncertainty, lags, and partly backward-looking expectations make real-world policy less clean than the pure RE benchmark.

Appendix: Policy Ineffectiveness Derivation

Under old-information policy:

$$\widehat{y}_t = v_t - \alpha_2 h(\pi^e_{t,t-1} - \pi^*) - \alpha_2 b(y^e_{t,t-1} - \bar{y}) - \alpha_2 \widehat{\rho}_t.$$

Rational expectations imply

$$y^e_{t,t-1} = E(y_t \mid I_{t-1}), \qquad \pi^e_{t,t-1} = E(\pi_t \mid I_{t-1}).$$

With zero expected shocks, the expectation equations give

$$y^e_{t,t-1} = \bar{y}, \qquad \pi^e_{t,t-1} = \pi^*.$$

Insert these back:

$$\widehat{y}_t = v_t - \alpha_2 \widehat{\rho}_t, \qquad \widehat{\pi}_t = \gamma v_t + s_t - \alpha_2 \gamma \widehat{\rho}_t.$$

The systematic policy coefficients disappear. Back

Appendix: Current-Information Policy and the Supply-Shock Trade-off

For simplicity set $b = 0$ and consider a supply shock $s_t > 0$ with no demand shock.

$$\widehat{y}_t = -\alpha_2 h \widehat{\pi}_t, \qquad \widehat{\pi}_t = \gamma \widehat{y}_t + s_t.$$

Substitute the first equation into the second:

$$\widehat{\pi}_t = \frac{s_t}{1 + \gamma\alpha_2 h}, \qquad \widehat{y}_t = -\frac{\alpha_2 h s_t}{1 + \gamma\alpha_2 h}.$$

A larger h lowers inflation but increases the negative output gap.

Back to PIP   Next appendix